

Week10HW-checkpoint

June 1, 2025

1 Week 10 Assignment

1.1 Problem 1

In the cell below, convert the string to all uppercase characters.

```
[1]: problem1_string = "Problem 1 isn't too bad!"  
     problem1_string.upper()
```

```
[1]: "PROBLEM 1 ISN'T TOO BAD!"
```

1.2 Problem 2

In the cell below, use python to find the first occurrence of the letter f in the string.

```
[2]: problem2_string = "supercalifragilisticexpialidocious"  
     letter = 'f'  
  
     # Find the first occurrence of the letter in the string  
     index = problem2_string.find(letter)  
  
     # condition of index being greater than 0  
     if index != -1:  
         print(f"The first occurrence of '{letter}' is at index {index}.")  
     else:  
         print(f"The letter '{letter}' is not found in the string.")
```

The first occurrence of 'f' is at index 9.

1.3 Problem 3

In the cell below, use the split function to separate the string at the commas and then print the value as a number.

```
[3]: problem2_string = "3,135,987,654,321,000"  
  
     split_problem2_string = problem2_string.split(',')  
     print(split_problem2_string)
```

```
['3', '135', '987', '654', '321', '000']
```

1.4 Problem 4

In the cell below load in the VA_Vote data and remove all duplicate rows.

```
[10]: # Imports
import matplotlib.pyplot as py
import pandas as pd

# Step 1: Read the data
VA_VOTES = pd.read_csv("VA_VOTES.csv")

# Step 2: Remove duplicates
cleaned_data = VA_VOTES.drop_duplicates()

# Step 3: Display cleaned data
print("Data after removing duplicates:")
print(cleaned_data)

# Optional: Save the cleaned data to a new CSV
cleaned_data.to_csv("cleaned_data.csv", index=False)
print("Cleaned data saved to 'cleaned_data.csv'.")
```

Data after removing duplicates:

	BLOCKID	Population	Voters	D_GOV	R_GOV	D_Percent	R_Percent
0	5200653	89.0	4.0	4.0	0.0	100.00%	0.00%
1	5200368	159.0	67.0	45.0	22.0	67.16%	32.84%
2	5200971	4.0	0.0	0.0	0.0	NaN	NaN
3	5200269	112.0	77.0	15.0	62.0	19.48%	80.52%
4	5200516	72.0	62.0	3.0	59.0	4.84%	95.16%
..	...	...	...	...	...	...	...
95	5200939	54.0	52.0	25.0	27.0	48.08%	51.92%
96	5200779	364.0	131.0	63.0	68.0	48.09%	51.91%
97	5200929	284.0	240.0	66.0	180.0	25.00%	75.00%
98	5200913	66.0	53.0	13.0	40.0	24.53%	75.47%
99	5200923	16.0	4.0	1.0	3.0	25.00%	75.00%

[99 rows x 7 columns]

Cleaned data saved to 'cleaned_data.csv'.

1.5 Problem 5

The cells below contain code that generates errors when run. Write a brief description of the problem and modify the code so that the cells execute properly.

```
[ ]: # Vert simple mistake the print statement needed to be indented
for i in range(10):
    print(i)
```

0
1
2
3
4
5
6
7
8
9

```
[ ]: # Assuming we want to print the values in the array we cannot use the function
      ↳Range
      # as it does not take in arguments and in order to print out the numbers x is
      ↳better suited
      for x in [3, 6, 12] :
          print(x)
```

3
6
12

```
[ ]: # There was some indenting problems
      # In python we do not use else if instead it is elif
      # we needed an action for 1 < 3.

      if 1 < 3:
          print("1 is indeed less than 3")
      elif 12 < 124:
          print("Math still works!")
```

1 is indeed less than 3